Facial Emotion Recognition: MLP vs CNN and the Role of Label Quality

A Comparative Study of Hard and Soft Supervision on FER-2013 and FER+

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Abstract

This report investigates two questions central to practical facial emotion recognition (FER): does model architecture or training label quality have greater impact on classification performance? Using the FER-2013 benchmark and its crowdsourced relabeling FER+, a Multi-Layer Perceptron (MLP) and a Convolutional Neural Network (CNN) were trained and compared under both hard single-annotator labels and soft probability distributions derived from ten annotators per image. The MLP plateaued at approximately 50% test accuracy on FER-2013 regardless of architectural tuning across 15 experimental runs (5 variants, 3 seeds each), while the CNN reached 66.84%—an 18.0 percentage point advantage attributable purely to spatial inductive bias. When soft labels from FER+ replaced the noisy hard labels, the same MLP architecture reached 66.70%, essentially matching the FER-2013 CNN (66.84%) despite having three times more parameters and fundamentally weaker spatial feature extraction. The combined CNN+FER+ configuration achieved 82.39%. Label quality improvement (+17.9pp for MLP) approximately equals architectural improvement (+18.0pp), and both effects compound significantly when combined. A noteworthy counter-finding is that minority classes (disgust, fear) regressed under soft label supervision due to near-uniform annotator vote distributions for rare, ambiguous classes, limiting FER+ gains to majority classes and highlighting an important design consideration for crowdsourced annotation strategies.

Keywords: facial emotion recognition, convolutional neural network, multi-layer perceptron, soft labels, label quality, FER-2013, FER+, soft cross-entropy, knowledge distillation, inductive bias

1 Introduction

Facial emotion recognition (FER) is the automated classification of human emotional states from facial images, with applications spanning human-computer interaction, driver distraction monitoring, pain assessment, and affective computing. Despite decades of research and the availability of large benchmark datasets, FER remains a genuinely difficult problem. A key reason is that the category boundaries between emotions are not sharp in pixel space: fear and sadness both produce furrowed brows and a downturned mouth, disgust and contempt share a lip curl, and neutral expressions can easily be mistaken for mild sadness [7]. This visual ambiguity is compounded by the fact that even trained human raters disagree substantially when labeling the same images: inter-annotator agreement on FER-2013 is only around 65% [2], meaning a significant fraction of the accuracy ceiling is perception-limited rather than model-limited.

The FER-2013 dataset [2], introduced at ICML 2013, has been the dominant benchmark for this task. It contains 35,887 web-scraped face images labeled by individual mechanical turk workers—one annotator per image with no quality control mechanism. Published accuracy on this benchmark consistently saturates around 65–68% for standard convolutional architectures, with marginal gains from increasingly complex models [7, 9]. This raises an important and underexplored question: how much of this ceiling is attributable to the noisy single-annotator labels, and how much to the fundamental difficulty of the visual task?

Microsoft Research addressed the label quality problem with FER+ [1], which relabels the same FER-2013 images using ten crowdsourced annotators per image. Rather than a single hard label, each image is assigned a full probability distribution over emotion categories, encoding both the majority opinion and genuine annotator uncertainty. Soft label training, where model outputs are supervised against probability distributions rather than one-hot vectors, has a well-established theoretical basis in knowledge distillation [5] and has been shown to improve calibration and generalization in various classification tasks.

In the context of FER specifically, Barsoum et al. [1] reported accuracy gains of 5–8 percentage points from soft label training on identical architectures, attributing the improvement to reduced gradient noise from contradictory labels. However, the relative magnitude of label quality effects compared to architectural improvements has not been systematically characterized.

This project was designed around the following research question: *If one could improve FER accuracy by ten percentage points, would that be better achieved through a more powerful model architecture or through cleaner training labels?* To answer this, an MLP and a CNN—representing fundamentally different architectural philosophies—were trained under both FER-2013 hard labels and FER+ soft labels, with architectures held constant across label conditions. This enables clean attribution of accuracy differences to either architectural inductive bias or label quality, without confounding factors.

The rest of this report is organized as follows. Section 2 describes the datasets, model architectures, and training procedures in detail. Section 3 presents and interprets the experimental results. Section 4 summarizes findings, discusses limitations, and outlines future directions.

2 Materials and Methods

2.1 Datasets

FER-2013. FER-2013 [2] contains 35,887 grayscale 48×48 face images across seven emotion classes: angry, disgust, fear, happy, neutral, sad, and surprise. The images were assembled from Google Image searches using emotion-related keywords and labeled by individual mechanical turk workers with no inter-rater reliability verification. The dataset provides three official splits: Training (28,709), PublicTest (3,589), and PrivateTest (3,574).

The class distribution is severely imbalanced: disgust has only 436 training examples while happy has 7,215—a ratio of approximately 1:17. This imbalance was addressed using class-weighted cross-entropy loss, where each class weight is inversely proportional to its training frequency. For the experimental split, the full training folder (28,709 images) was used for model training, and the official test folder (7,178 images) was divided evenly into a validation set and a held-out test set of 3,589 images each using a fixed random seed of 42. This approach preserves the entire training set for learning while providing a clean validation/test split from the same test distribution.

FER+. FER+ [1] relabels the same FER-2013 images with ten crowdsourced annotators per image, producing a probability vector over eight emotion categories (the original seven plus contempt), along with "unknown" and "non-face" (NF) response counts. The soft label representation captures genuine annotator uncertainty in a way that a single hard label fundamentally cannot.

After removing non-face images (three images with NF votes $> 50\%$) and images where contempt was the plurality label (221 images, 0.6% of the dataset), 35,490 usable seven-class samples remained. The official Training (28,392 usable), PublicTest (3,554 as validation), and PrivateTest (3,544 as test) splits were used.

Figure 1 shows the label quality analysis. The mean label entropy was 0.718 bits (theoretical maximum 3.0 bits for a uniform eight-class distribution), indicating that the majority of images have a clear dominant label. Approximately 54.9% of usable samples had annotator confidence $\geq 80\%$, while 6.7% were ambiguous with confidence below 50%. The 26,248 low-entropy samples (below 1.0 bit) are well-approximated by near-one-hot distributions, while the 479 high-entropy samples (above 2.0 bits) represent genuinely ambiguous cases where the soft label treatment provides the most benefit during training.

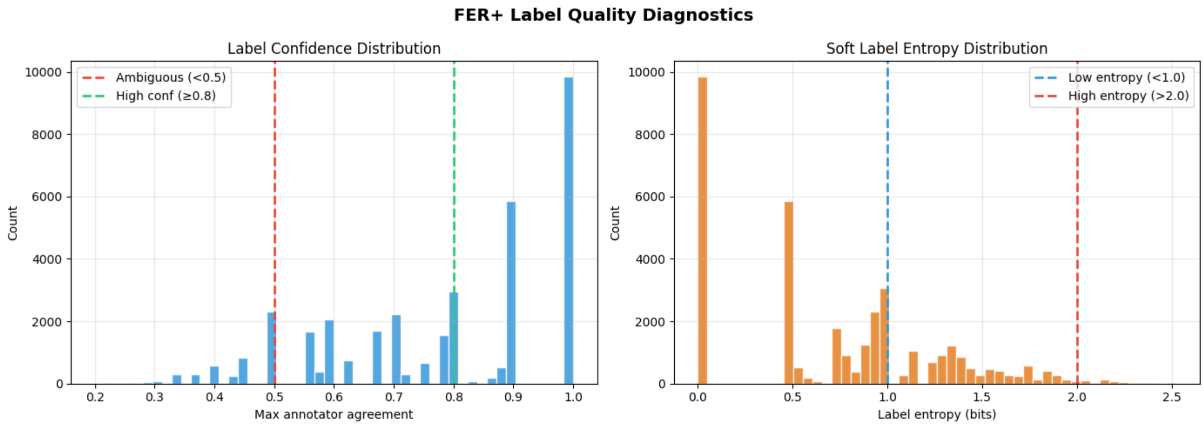


Figure 1: FER+ label quality diagnostics. **Left:** distribution of maximum annotator agreement (confidence). The large spike at 1.0 indicates images where all ten annotators agreed unanimously. **Right:** soft label entropy distribution. The majority of images have entropy below 1.0 bit, confirming that soft labels are informative rather than uninformative for most of the dataset. The 6.7% of samples with confidence below 0.5 represent genuinely ambiguous face images.

2.2 Model Architectures

MLP (wider_deeper). The Multi-Layer Perceptron flattens each 48×48 input image into a vector of 2,304 pixel values. The best-performing variant, *wider_deeper*, consists of four fully connected hidden layers with dimensions [2048, 1024, 512, 256], each followed by batch normalization, ReLU activation, and dropout. Progressive dropout rates [0.35, 0.30, 0.25, 0.20] decrease with depth, reflecting the intuition that deeper layers encode more abstract and stable representations that require less regularization. The output layer maps to seven classes. Total parameters: 7,484,423.

The fundamental architectural limitation of the MLP is the input flattening operation. Pixel (0,0) and pixel (47,47)—diagonally opposite corners of the face—appear in the input as equally related as adjacent pixels (23,23) and (24,23). The model therefore cannot learn that the relative positions of facial landmarks (eyes above nose above mouth) are informative for emotion classification. This prior knowledge is trivially true for face images but is simply not representable in the MLP's weight structure. Importantly, this limitation cannot be addressed by making the network wider or deeper; it is a structural consequence of the flattening operation [6].

CNN (baseline). The Convolutional Neural Network consists of four convolutional blocks with channel counts [32, 64, 128, 256]. Each block contains two Conv3×3 layers, batch normalization, and ReLU activations, followed by MaxPool2d(2) spatial downsampling and Dropout2d(0.25) channel-wise regularization. Four MaxPool2d operations reduce the spatial dimensions from 48×48 to 3×3. The resulting 256×3×3 feature map is flattened and processed by a two-layer classifier with 512 hidden units and 50% dropout. Total parameters: 2,358,375—approximately three times fewer than the MLP.

The CNN's advantage comes from weight sharing: the same 3×3 filter is applied at every spatial location, requiring only nine parameters per filter to detect a local pattern anywhere in the image. This structural prior—that local pixel relationships are more informative than global ones—is known as an inductive bias [6, 4] and is well-matched to face images where the spatial arrangement of features like eye corners, nose bridge, and mouth curvature carries the diagnostic information for emotion classification.

2.3 Loss Functions

For FER-2013 experiments, class-weighted cross-entropy with label smoothing ($\epsilon = 0.1$) was used:

$$\mathcal{L}_{\text{CE}} = -\sum_{c=1}^{C} w_c \left[(1-\epsilon)y_c + \frac{\epsilon}{C}\right] \log \hat{p}_c \tag{1}$$

where w_c are class weights inversely proportional to class frequency and $y_c \in \{0, 1\}$ is the hard label. For FER+ experiments, soft cross-entropy was used:

$$\mathcal{L}_{\text{Soft}} = -\sum_{c=1}^{C} p_c^{\text{target}} \log \hat{p}_c^{\text{model}} \tag{2}$$

where p_c^{target} is the normalized annotator vote distribution. No class weights or label smoothing were required for FER+, because the soft targets implicitly encode class frequency through the vote distributions and already capture uncertainty information that label smoothing approximates artificially.

2.4 Training Configuration

Both models used AdamW optimization [8] with a learning rate schedule consisting of linear warmup followed by cosine annealing to a minimum of 10^{-6}. Gradient clipping with maximum norm 1.0 was applied throughout. Table 1 summarizes key hyperparameters.

Table 1: Training configuration for all experiments.

Setting	MLP (FER-2013)	CNN (FER-2013)	FER+ (both)
Learning rate	10^{-3}	3×10^{-4}	same
Weight decay	5×10^{-5}	10^{-4}	same
Batch size	128	64	128
LR warmup	5 epochs	10 epochs	same
Max epochs	100	120	120/150
Early stop patience	10	10	20
Loss	CE + class weights	CE + class weights	Soft CE
Label smoothing	0.1	0.1	—
Gradient clipping	1.0	1.0	1.0

The larger patience (20 epochs) for FER+ experiments reflects an empirically observed property of soft CE training: validation accuracy continues improving slowly throughout training due to the smooth gradient landscape of probability target supervision. Using patience=10 (as in FER-2013) would terminate FER+ training prematurely, as confirmed by the MLP running its full 120-epoch budget without early stopping.

2.5 Experimental Protocol

For FER-2013, five MLP variants were trained with three random seeds each (15 total runs): *baseline* [1024, 512, 256] at lr=10^{-3}; *wider_deeper* [2048, 1024, 512, 256] at lr=10^{-3}; *lower_lr* at lr=5×10^{-4}; *high_dropout* with stronger per-layer dropout; and *high_wd* with increased weight decay 10^{-4}. The *wider_deeper* variant achieved the best mean test accuracy and was selected for FER+ comparison. The CNN baseline was trained with three seeds. For FER+ experiments, one run per architecture using seed=42 was conducted.

Model selection was performed exclusively on the validation set; the test set was evaluated once at the end using the checkpoint from the epoch of highest validation accuracy. The test set was never used for any selection decision.

3 Results and Evaluation

3.1 MLP on FER-2013: Representational Ceiling

Table 2 shows results for the three primary MLP variants. All converged near 50% test accuracy regardless of configuration. The wider_deeper model, with 7.5M parameters versus the baseline's 3M, gained only 0.86pp. The lower_lr variant achieved essentially identical accuracy with substantially lower variance (std ±0.06%), suggesting reliable convergence to the same solution across seeds.

The seed=42 wider_deeper run reached 48.82% test accuracy (best val ep=79, early stop ep=89). Per-class results: happy 62.4%, disgust 62.5%, surprise 65.5%, neutral 48.7%, angry 40.9%, sad 38.0%, fear

Table 2: MLP variant results on FER-2013 (3 seeds each, 50/50 test split, seed=42 selected).

Variant	Architecture	Mean Test Acc	Std
baseline	[1024, 512, 256]	49.31%	±0.47%
wider_deeper	[2048, 1024, 512, 256]	50.17%	±0.27%
lower_lr	[1024, 512, 256]	49.13%	±0.06%

31.8%. The two hardest classes—fear and sadness—share furrowed brows and a downturned mouth, making them visually nearly indistinguishable at 48×48 resolution [7].

Figure 2 (top row) shows the training curves. Validation accuracy stabilizes around 50% while training accuracy continues rising, confirming that the bottleneck is representational rather than a classical bias-variance overfitting problem. The large train-validation gap (approximately 10pp) is diagnostic of a model that has memorized training pixel patterns but cannot generalize because those patterns do not transfer to the test distribution without spatial structure.

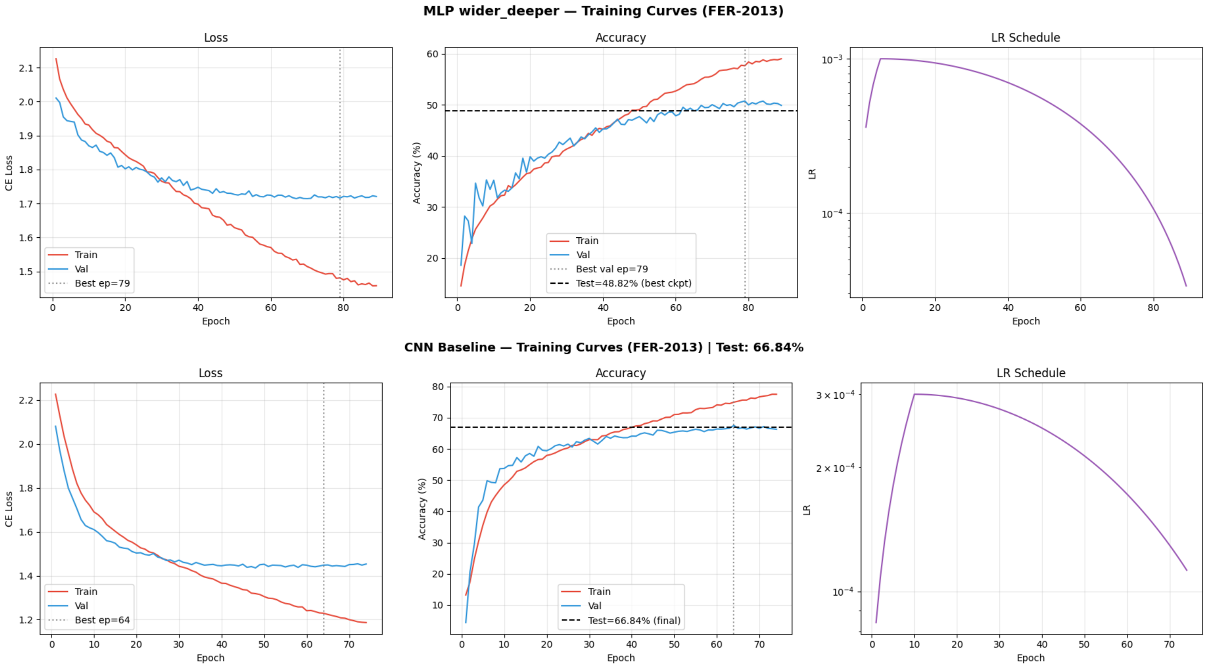
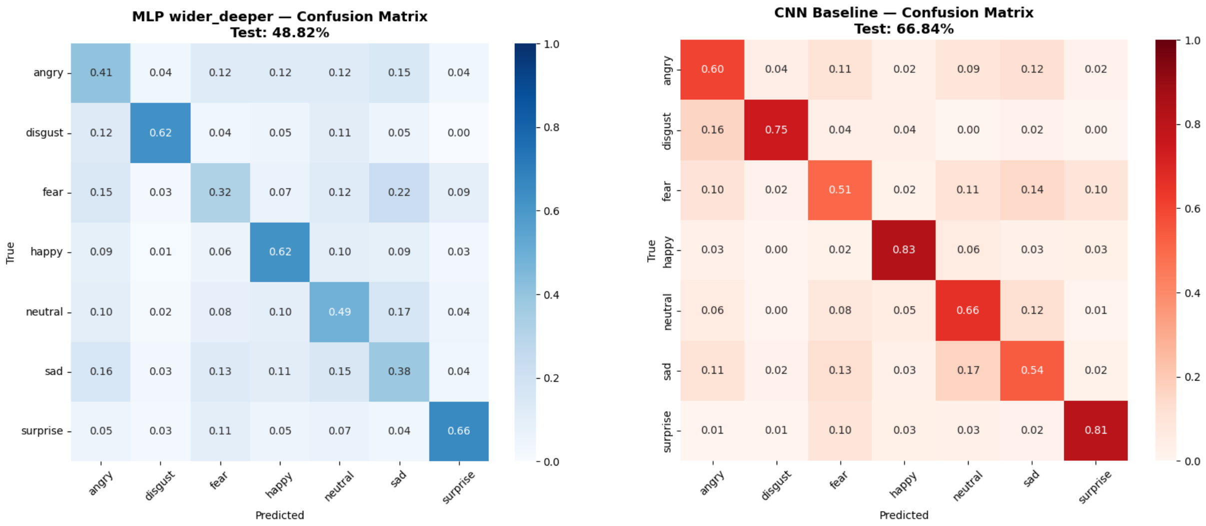


Figure 2: Training curves for FER-2013. **Top:** MLP wider_deeper (48.82%, best ep=79). Validation stabilizes around 50% while training continues rising—a representational ceiling, not classical overfitting. The cosine LR schedule is visible in the third panel. **Bottom:** CNN baseline (66.84%, best ep=64). The val curve converges cleanly and the single test evaluation at the best checkpoint confirms strong spatial feature generalization.

3.2 CNN on FER-2013: Spatial Feature Learning

The CNN reached 66.84% test accuracy (best val ep=64, early stop ep=74)—an 18.0pp improvement over the MLP using three times fewer parameters. Figure 3 shows the side-by-side normalized confusion matrices.

The CNN diagonal is substantially darker across most classes: disgust improves from 0.62 to 0.75, neutral from 0.49 to 0.66, and angry from 0.41 to 0.60. The improvement on disgust is particularly notable given the small sample count (approximately 56 test images in the 50/50 split): the CNN learns a reliable spatial signature (nose wrinkle, lip curl) that the MLP cannot represent. Both models continue to struggle on fear (CNN: 0.51, MLP: 0.32), and sadness remains confused with neutral for both (CNN: 0.17 sadness → neutral; MLP: 0.15), reflecting genuine visual overlap rather than a model limitation.



(a) MLP wider_deeper: 48.82%

(b) CNN baseline: 66.84%

Figure 3: Normalized confusion matrices on the FER-2013 test set (50/50 split). The CNN diagonal is substantially darker across most classes. Fear remains the hardest class for both models. The sadness → neutral confusion in both matrices reflects genuine visual ambiguity at 48×48 resolution rather than a model limitation.

3.3 FER+ Soft Labels: Label Quality Effect

Figure 4 shows the training curves for both models under FER+ supervision. The MLP reached 66.70% (best val ep=117, all 120 epochs used—early stopping never triggered), and the CNN reached 82.39% (best val ep=88, early stop ep=108). Two qualitative differences from FER-2013 training are immediately visible.

First, for the MLP, validation accuracy continues improving slowly until epoch 117 without plateauing. This is a property of soft cross-entropy supervision: because targets are probability distributions rather than delta functions, the gradient signal remains non-zero even at low learning rates, allowing the cosine LR schedule's long tail to extract further improvements. Under hard CE, the loss saturates once the model confidently predicts the correct hard label; under soft CE, the loss continues to distinguish better from worse probability estimates throughout training.

Second, for the CNN, training and validation accuracy converge to within 4–5pp of each other by the end of training, compared with a 15+ point train-val gap in FER-2013 CNN training. This is because soft labels act as an implicit regularizer: they prevent the model from becoming overconfident, and the consensus distributions for majority-class images provide consistent, non-contradictory gradient updates across repeated exposures to the same image.

Figure 5 shows the FER+ confusion matrices. The CNN diagonal is strong for majority classes: happiness 0.91, surprise 0.88, neutral 0.87, and anger 0.79. Sadness improved substantially from 0.54 (FER-2013 CNN) to 0.60, suggesting that many sadness images were mislabeled in FER-2013 and the FER+ consensus revealed their correct category. The disgust and fear rows, however, show significant off-diagonal mass in both matrices. Disgust (23 PrivateTest samples) has only 39% correctly classified by the CNN, with 30% predicted as anger and 17% as sadness; fear (94 samples) has only 40% correctly classified with 36% predicted as surprise.

The per-class FER+ MLP results reveal a strongly bimodal pattern. Majority classes improved substantially: neutral +30.6pp (48.7% → 79.3%) and happiness +14.1pp (62.4% → 76.5%) over FER-2013 MLP. Anger showed a modest gain (+4.5pp, from 40.9% to 45.4%) and surprise improved slightly (+2.9pp, from 65.5% to 68.4%), consistent with both being visually distinct classes that benefit from cleaner supervision even if they were already reasonably well-classified. Minority classes regressed sharply: disgust −40.8pp (62.5% → 21.7%) and fear −10.5pp (31.8% → 21.3%). This regression has a clear mechanism. Disgust has 436 training samples, many of which received highly split annotator votes in FER+. For these images, the normalized vote distribution is near-uniform across several emotion categories. Training against a

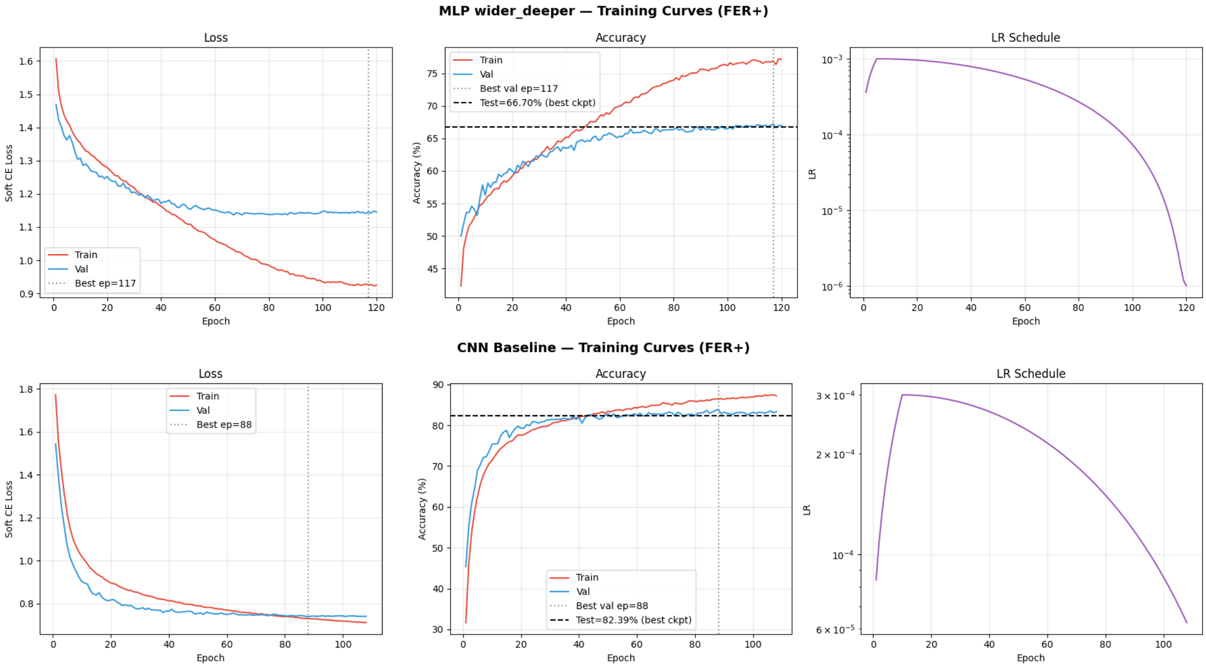


Figure 4: Training curves for FER+ experiments. **Top:** MLP wider_deeper (66.70%, best ep=117). No early plateau occurs—soft CE keeps providing gradient signal for the full 120-epoch budget. Test is reported as a single horizontal line at the best checkpoint value. **Bottom:** CNN baseline (82.39%, best ep=88, early stop ep=108). Train and val converge closely, indicating clean learning from soft labels without the large train-val gap seen in FER-2013 training.

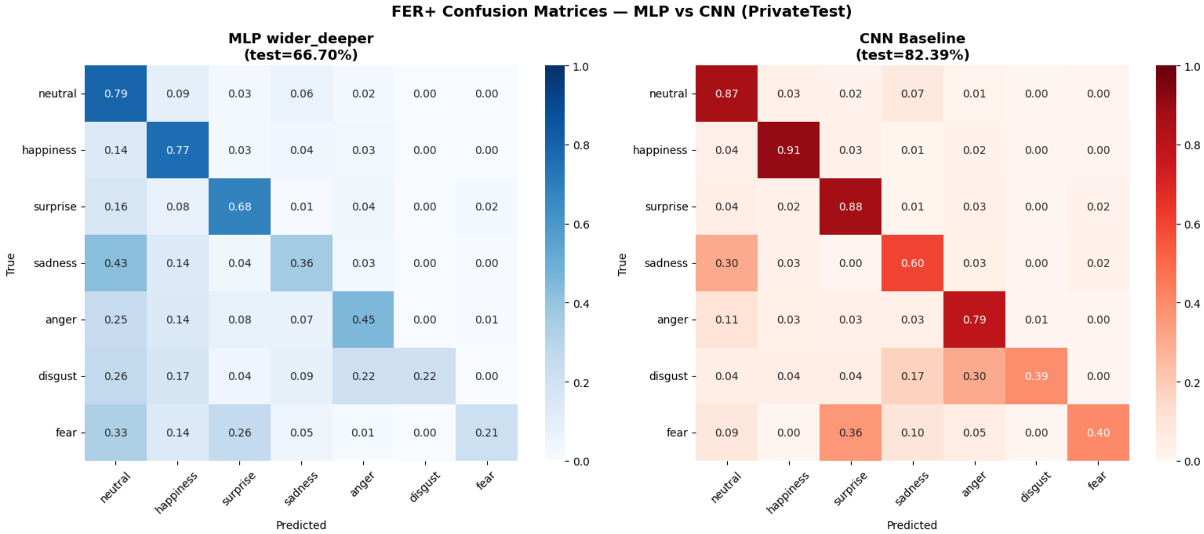


Figure 5: Normalized FER+ confusion matrices on PrivateTest. **Left:** MLP 66.70%. **Right:** CNN 82.39%. Majority classes (happiness, surprise, neutral, anger) are well-classified. Disgust and fear show significant off-diagonal scatter, reflecting ambiguous soft label distributions for these rare classes—when annotators disagree, the soft target is near-uniform and provides weak gradient signal.

near-uniform target provides almost no directional gradient information: the model cannot learn which class to prefer, and effectively unlearns whatever disgust features were acquired from hard labels. This is precisely the scenario in which soft labels provide weaker supervision than noisy hard labels.

The CNN under FER+ shows the same bimodal pattern: strong majority-class improvements (neutral +20.3pp: 66.4% → 86.7%, anger +18.9pp: 60.2% → 79.1%, surprise +6.5pp: 81.2% → 87.7%, happiness +8.1pp: 82.9% → 91.0%, sadness +7.0pp: 53.5% → 60.5%) alongside regression in disgust (−35.9pp: 75.0% → 39.1%) and fear (−10.2pp: 50.6% → 40.4%), mirroring the mechanism described above.

3.4 Overall Comparison and Discussion

Table 3 and Figure 6 summarize the four canonical experimental results. The key quantitative comparisons are:

- **Architecture gap (FER-2013):** CNN vs MLP = +18.0pp, achieved with 3× fewer parameters
- **Label quality gain (MLP):** FER+ vs FER-2013 = +17.9pp, same architecture
- **Label quality gain (CNN):** FER+ vs FER-2013 = +15.6pp, same architecture
- **Architecture gap (FER+):** CNN vs MLP = +15.7pp (slightly smaller than FER-2013 because the MLP benefits more proportionally from soft labels)

Table 3: Summary of canonical results across all four experiments.

Experiment	Test Acc	Best Val Ep	Val Acc	Δ from MLP baseline
FER-2013 MLP (hard labels)	48.82%	79	50.74%	—
FER-2013 CNN (hard labels)	66.84%	64	67.51%	+18.0pp
FER+ MLP (soft labels)	66.70%	117	67.22%	+17.9pp
FER+ CNN (soft labels)	**82.39%**	88	83.82%	+33.6pp

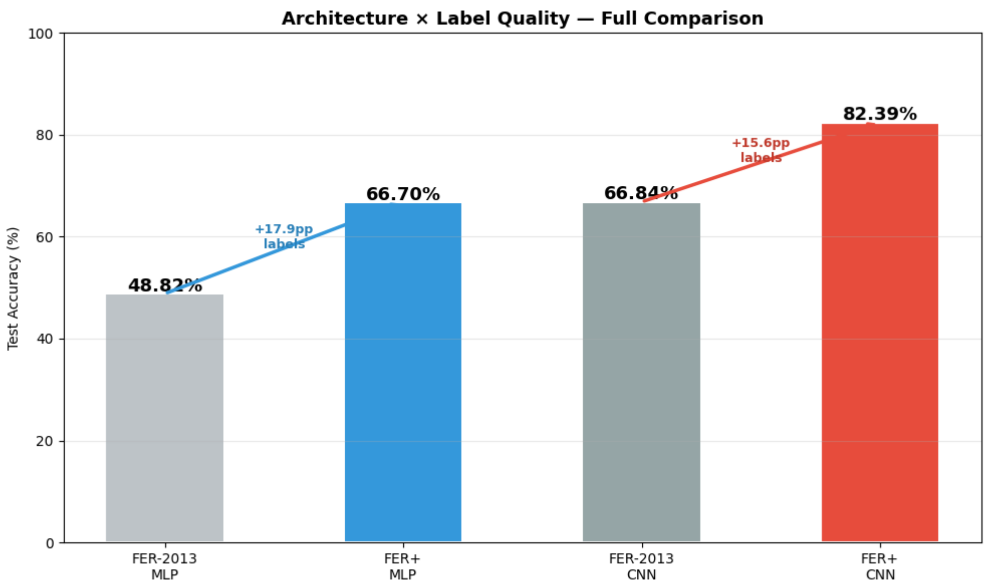


Figure 6: Architecture × Label Quality: full comparison across all four experiments. Label quality gain (MLP: +17.9pp, CNN: +15.6pp) is approximately equal to the architecture gap (+18.0pp). Most significantly, FER+ MLP (66.70%) and FER-2013 CNN (66.84%) are within 0.14pp of each other—within single-seed variance—demonstrating that clean supervision on a structurally simpler model can match a more powerful model trained on noisy labels.

The most practically significant finding is that the label quality gain (≈+18pp) approximately equals the architecture gain (+18pp) in absolute terms. The near-tie between FER+ MLP (66.70%) and FER-2013 CNN (66.84%) is the clearest demonstration of this: both differ by only 0.14pp, well within the variance expected from a single seed. This result has an important implication for practitioners: in settings where engineering a better architecture is costly or requires domain expertise, investing in annotation quality—collecting more raters per image or using crowd consensus protocols—can achieve comparable accuracy improvements. The caveat, demonstrated by the disgust and fear regression, is that soft labels work best when the annotator pool is large enough and consistent enough to produce informative distributions for the rarest and most ambiguous categories in the dataset.

4 Conclusion

This project systematically compared the contribution of model architecture and training label quality to facial emotion recognition performance. The headline finding is that both effects are approximately equal in magnitude for the FER-2013/FER+ setting (+18.0pp from CNN architecture, +17.9pp from FER+ soft labels on the MLP) and that they compound substantially when applied together: CNN+FER+ achieves 82.39% from the MLP+FER-2013 baseline of 48.82%—a 33.6pp total improvement.

The central and most practically useful finding is that FER+ MLP (66.70%) and FER-2013 CNN (66.84%) are within 0.14pp of each other. A label quality improvement on a structurally simpler model can match a better-designed model trained on noisier labels. For practitioners who cannot easily redesign their model architecture, this result suggests that re-annotation campaigns using crowd consensus protocols represent a high-leverage intervention of comparable magnitude to an architectural upgrade.

The minority class regression under soft labels is the key counter-finding, and it has practical design implications. Disgust regressed by 35.9 percentage points on the CNN (75.0% $\rightarrow$ 39.1%) and by 40.8 points on the MLP (62.5% $\rightarrow$ 21.7%) under FER+ supervision. Fear fell by 10.2 points on the CNN (50.6% $\rightarrow$ 40.4%) and 10.5 points on the MLP (31.8% $\rightarrow$ 21.3%). This occurs because rare-class images with genuinely ambiguous visual features receive near-uniform soft targets, providing weaker gradient signal than even a noisy hard label. The conclusion is that soft labels are not universally beneficial: their effectiveness depends critically on the consistency of the annotator pool for each class. A hybrid strategy—using soft labels for well-represented classes and hard labels (or class-specific temperature scaling) for rare, ambiguous classes—is a promising direction for future work.

Limitations. FER+ experiments were conducted with a single random seed (seed=42), so no variance estimate is available for these results. The disgust class has only 23 PrivateTest samples, making its per-class accuracy highly sensitive to individual predictions. FER-2013 images are web-scraped and uncontrolled for pose variation, lighting conditions, occlusion, and JPEG compression, which set a ceiling on achievable accuracy independent of the supervision strategy.

Future work. Temperature scaling of soft targets could sharpen minority class distributions without abandoning uncertainty encoding for majority classes [3]. ResNet and Vision Transformer backbones trained on FER+ are reported at 85%+ accuracy in the literature [7], suggesting that further architectural improvements remain available once label quality is fixed. FER+ as a pre-training source for fine-tuning on in-the-wild datasets such as AffectNet [9] and RAF-DB represents a natural next step toward deployment-ready FER systems.

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